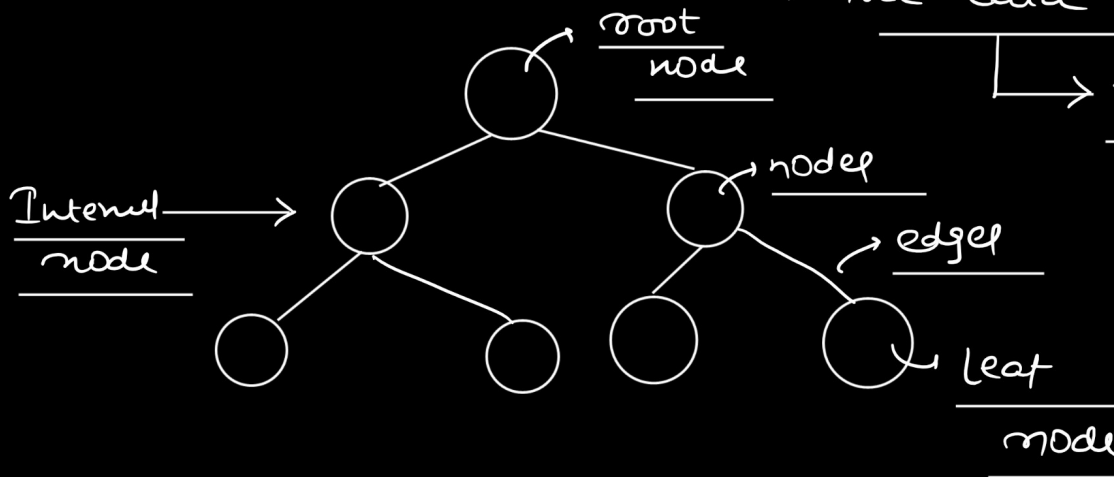


Decision Tree (Regression, Classification)

→ Tree data structure

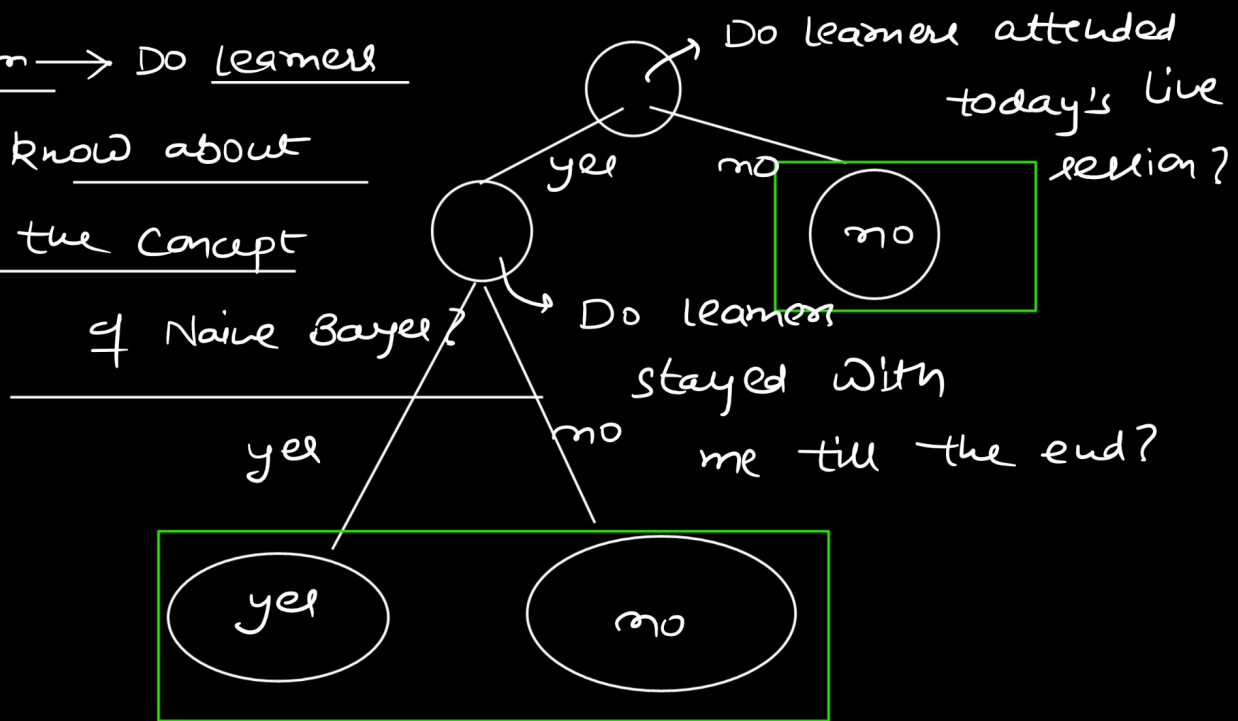
→ Non-linear



Problem → Do learners

know about
the concept

of Naive Bayes?



Gini Index & Entropy

→ Measure of Impurity

$$\underline{Gini} = 1 - \sum_{i=1}^c p_i^2$$

$Gini(class) \Rightarrow$

$Gini(Gender) \Rightarrow$

(F₁) ✓ class

Gender(F₁) stay in hostel

target

Given
to
you

9	M	Y
10	F	N
8	F	Y
8	F	N
9	M	Y
10	M	N
11	F	Y
11	M	Y
8	F	Y
9	M	N
11	M	N
11	M	Y
10	F	N
10	M	Y

⇒ 14 num of
records

$$1 - \sum_{i=1}^c p_i^2$$

Gini value

4/9

8

Y=2 N=1

2/3

1/3

9

Y=2 N=1

2/3

1/3

10

Y=1 N=3

1/4

3/4

11

Y=3 N=1

3/4

1/4

$$1 - \left(\left(\frac{1}{4} \right)^2 + \left(\frac{3}{4} \right)^2 \right) = \frac{3}{8}$$

Gini(class) → $\frac{3}{14} * \frac{4}{9} + \frac{3}{14} * \frac{4}{9} + \frac{4}{14} * \frac{3}{8} + \frac{4}{14} * \frac{3}{8}$

↳ Weighted
avg

$$\Rightarrow \underline{0.404 \text{ (minimum)}}$$

Gender	Stay in hostel	P(Y)	P(N)	Gini
M	Y=5 N=3	5/8	3/8	$1 - ((5/8)^2 + (3/8)^2)$
F	Y=3 N=3	3/6	3/6	$1 - ((3/6)^2 + (3/6)^2)$

$$Gini(Gender) = \frac{8}{14} \times 0.469 + \frac{6}{14} \times 0.5$$

$$\Rightarrow \underline{0.482}$$

